

Configuration and Usage Manual

A practical reference for configuring, running and installing the Bash tool for the Proton VPN command-line interface.

DOCUMENT EDITION • English
VARIABLE NAMES AND CONFIGURATION VALUES PRESERVED
DESIGN SYSTEM • #6d4aff

WHAT THIS MANUAL COVERS

Configuration reference Destination selection, route mode, switching intervals, Secure Core, notifications and retry behavior.	Operation and installation Direct execution, command installation, desktop autostart, exit codes and security notes.
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SCOPE
protonvpn-random controls the Proton VPN Linux CLI through the protonvpn command. It does not control the graphical Proton VPN desktop application.

Document map

The manual separates configuration, operation and security so each topic can be found quickly.

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READING PATH

New users should read sections 01 and 02 first. Experienced users can jump directly to section 03 for the configuration reference or section 04 for installation.

Overview and requirements

Understand what the script does, what it depends on and how the major controls interact.

Purpose

protonvpn-random is a small Bash tool for the Proton VPN command-line interface. It connects to a selected Proton VPN destination and can rotate the connection automatically after a randomized waiting period.

CLI ONLY

The script invokes the protonvpn command. It does not automate the Proton VPN graphical application.

Core features

- Preferred destination with a configurable probability in RANDOM mode
- Random pool selection or ordered traversal in ROUTE mode
- Randomized time window between connection changes
- Optional Secure Core probability
- Retry handling with increasing delays
- Granular desktop notifications and terminal output
- Configuration validation and defined exit codes
- Command-line options that can override matching script settings

Required commands

bash	Runs the script.
protonvpn	Controls the official Proton VPN Linux CLI.
notify-send	Displays desktop notifications.
tput	Formats terminal output and the countdown display.

Underlying Proton VPN CLI

The Proton VPN CLI must already be installed, configured and signed in. The current official Linux CLI uses NetworkManager and a compatible keyring and is not intended for headless systems.

Getting started

Run the script directly, inspect available options and understand which settings take precedence.

1. Make the script executable

```
chmod +x protonvpn-random
```

2. Start it

```
./protonvpn-random
```

Alternative invocation

```
bash protonvpn-random
```

3. Inspect command-line help

```
./protonvpn-random -h
```

Command-line precedence

Script defaults

>

User configuration

>

CLI options

>

Effective run

When a supported command-line option is supplied, it overrides the corresponding value stored in the script for that run. It does not permanently rewrite the configuration section.

TERMINAL BEHAVIOR

When started in a terminal, the script can show startup information and the remaining time until the next connection change. These displays are controlled by `PROTON_INFO_ONOFF` and `START_MESSAGE_ONOFF`.

First-run checklist

- The `protonvpn` command works and the account is signed in.
- The location identifiers are accepted by the installed CLI.
- `SECONDS_SWITCH_VPN_MIN` is not greater than `SECONDS_SWITCH_VPN_MAX`.
- Notification and terminal dependencies are available.

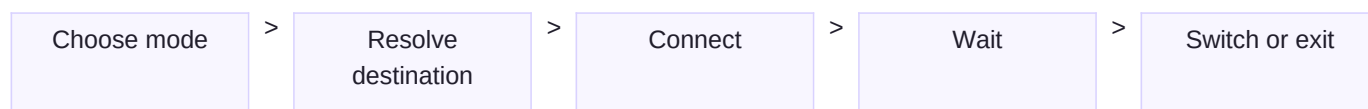
Configuration reference

The new MODE setting changes the structure: destination selection and traversal logic are documented separately.

Configuration groups

3.1	Identity and preferred destination - APPNAME, MAIN_LOCATION and probability
3.2	Pool and selection mode - RANDOM_POOL and MODE
3.3	Switching and timing - automatic rotation and minimum/maximum delay
3.4	Secure Core - probability and unsupported destination behavior
3.5	Notifications and terminal output - global and individual controls
3.6	Failure handling - retry limit and increasing delay

How destination selection works



RANDOM mode can choose MAIN_LOCATION according to PROBABILITY_MAIN_LOCATION; otherwise it chooses from RANDOM_POOL. ROUTE mode walks through RANDOM_POOL in its defined order. The switch delay remains randomized within the configured interval in both modes.

VALIDATION PRINCIPLE

Use valid Bash syntax and values from the documented ranges. In particular, MODE must use straight ASCII quotes: MODE="RANDOM". Typographic quotation marks are not valid Bash string delimiters.

Accepted destination forms

Country	US
City	Amsterdam, New York, Salt Lake City
Server	US-FREE#36, US-GA#29-TOR, NO#12

Identity and preferred destination

Define the notification identity and the preferred destination used by RANDOM mode.

APPNAME="\$(basename "\$0")"

The application name is displayed in notifications. Depending on the desktop environment, notification history may associate messages with this name. By default, the name matches the script filename.

MAIN_LOCATION="US"

Defines the preferred VPN destination for RANDOM mode. It may be a country, city or individual server identifier. When the MAIN_LOCATION probability condition is met, the script does not select an entry from RANDOM_POOL for that connection.

PROBABILITY_MAIN_LOCATION=0

Controls how often MAIN_LOCATION is selected in RANDOM mode.

ALLOWED VALUES 0 - 100

0	MAIN_LOCATION is not used; selection comes from RANDOM_POOL.
50	MAIN_LOCATION is selected with a probability of approximately 50 percent.
100	MAIN_LOCATION is always selected; RANDOM_POOL is not used in RANDOM mode.

Examples

Country	MAIN_LOCATION="US"
City	MAIN_LOCATION="New York"
Server	MAIN_LOCATION="US-FREE#36"

MODE SCOPE

The preferred-location probability is a RANDOM-mode feature. ROUTE mode follows the sequence defined in RANDOM_POOL.

Pool and selection mode

Define the available destinations and choose random selection or ordered traversal.

RANDOM_POOL

The destination pool used when MAIN_LOCATION is not selected in RANDOM mode. In ROUTE mode, the same pool defines the traversal order. Countries, cities and individual server identifiers may be mixed.

```
RANDOM_POOL=(
  "US-FREE#36"
  "US-GA#29-TOR"
  "New York"
  "NO"
)
```

MODE="RANDOM"

Selects how destinations from RANDOM_POOL are processed.

ALLOWED VALUES RANDOM, ROUTE

RANDOM

Pool entries are selected randomly. MAIN_LOCATION can be inserted according to PROBABILITY_MAIN_LOCATION.

ROUTE

Pool entries are processed one after another in their written order. The switch time is still randomized.

POOL DESIGN

Do not add MAIN_LOCATION to RANDOM_POOL unless you intentionally want to increase its total selection frequency in RANDOM mode.

Practical examples

Mixed pool	Countries, cities and servers can be combined.
Predictable route	Use ROUTE when the connection order matters.
Maximum variety	Use RANDOM with a diverse pool and a low main-location probability.

Switching and timing

Control whether rotation continues and define the random delay window between switches.

SWITCH_VPN_ONOFF="ON"

Controls whether the script continues changing the VPN connection after the first successful connection.

ALLOWED VALUES ON, OFF

ON

After the randomized waiting time, the script selects the next destination and starts another connection attempt.

OFF

The script establishes one connection and then exits. The VPN connection itself remains active.

SECONDS_SWITCH_VPN_MAX=10800

Maximum number of seconds before the next switch. This value must not be smaller than SECONDS_SWITCH_VPN_MIN.

ALLOWED VALUES integer

SECONDS_SWITCH_VPN_MIN=600

Minimum number of seconds before the next switch. This value must not be greater than SECONDS_SWITCH_VPN_MAX.

ALLOWED VALUES integer

Timing model



For example, MIN=3600 and MAX=7200 produce a randomized waiting time between one and two hours.

VALIDATION

The script should reject a timing range where the minimum exceeds the maximum. Use non-negative integer values that match the intended operating interval.

Secure Core probability

Control how often the script requests a Secure Core connection and understand compatibility failures.

PROBABILITY_SECURECORE=0

Controls the probability that a connection attempt uses Secure Core.

ALLOWED VALUES 0 - 100

0	Secure Core is disabled.
50	Secure Core is requested for approximately half of the connection attempts.
100	Secure Core is requested for every connection attempt.

Compatibility behavior

Not every destination type or individual server supports Secure Core. A probability greater than zero can therefore produce failed connection attempts when the chosen pool entry is incompatible.



RECOVERY BEHAVIOR

When a Secure Core attempt fails because the selected destination is unsupported, protonvpn-random reports the failure and proceeds according to its retry and selection logic.

Configuration guidance

- Use country-level entries when Secure Core compatibility is important.
- Test server and city identifiers before relying on them in unattended operation.
- Set the probability to 0 when deterministic compatibility is more important than Secure Core selection.

Notifications and terminal output

Configure global notification muting, individual messages and terminal-only information.

NOTIFY_ALLOFF="NO"

Master control for desktop notifications. Setting this variable to OFF disables the individual notification categories below. Any other supported state leaves them individually configurable.

ALLOWED VALUES NO, OFF

NOTIFY_VPN_ERROR_ONOFF="ON" NOTIFY_VPN_OK_ONOFF="ON" NOTIFY_EXIT_ONOFF="ON" NOTIFY_ASK_ONOFF="ON"

Individual controls for error, successful connection, exit and confirmation/request notifications.

ALLOWED VALUES ON, OFF

PROTON_INFO_ONOFF="ON"

Loads and displays output from protonvpn at startup when protonvpn-random is running in a terminal.

ALLOWED VALUES ON, OFF

START_MESSAGE_ONOFF="ON"

Displays a short introductory message when protonvpn-random is running in a terminal.

ALLOWED VALUES ON, OFF

Control hierarchy

NOTIFY_ALLOFF

>

Individual switches

>

notify-send

BACKGROUND MODE

Terminal-only information is naturally invisible when the program is started without a terminal. Desktop notifications can still appear when the graphical session and notification service are available.

Failure handling

Define how many consecutive connection failures are tolerated and how the retry delay grows.

MAX_CON_FAILS=2

Maximum number of consecutive connection failures allowed before protonvpn-random exits. Exiting the script does not automatically disconnect a VPN connection that was already active.

ALLOWED VALUES integer

INCREASING_DELAY=10

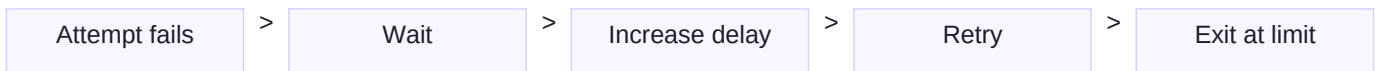
Base number of seconds added to the waiting time after each failed connection attempt. The delay grows by this same amount with every consecutive failure.

ALLOWED VALUES integer

Example with INCREASING_DELAY=10

Failure 1	Wait 10 seconds before the next attempt.
Failure 2	Wait 20 seconds before the next attempt.
Failure 3	Wait 30 seconds before the next attempt.

Retry flow



EXISTING CONNECTION

A script exit after repeated failures does not by itself prove that an existing VPN tunnel was disconnected. Check the current Proton VPN status separately when that distinction matters.

Install as a command

Place the script in `/usr/local/bin` so it can be started from any working directory.

Copy and set permissions

```
sudo cp protonvpn-random /usr/local/bin/protonvpn-randomizer
sudo chmod 755 /usr/local/bin/protonvpn-randomizer
```

Start it from anywhere

```
protonvpn-randomizer
```

Edit the installed configuration

```
sudo nano /usr/local/bin/protonvpn-randomizer
```

CONFIGURATION LOCATION

The user configuration remains embedded in the copied script. Editing the original file does not update the installed copy automatically.

Update workflow

1	Keep a backup of the installed configuration.
2	Replace the installed script with the newer version.
3	Reapply the intended configuration values.
4	Run the help and configuration validation before unattended use.

Alternative name

The example installs the command as `protonvpn-randomizer` to distinguish the installed command from the source filename. You may keep the original name instead, provided all paths and autostart entries use the same command.

Start automatically after desktop login

Create a desktop autostart entry and choose whether a terminal window should be visible.

Create the autostart directory

```
mkdir -p "$HOME/.config/autostart"
```

Create this file

```
$HOME/.config/autostart/protonvpn-randomizer.desktop
```

Example desktop entry

```
[Desktop Entry]
Type=Application
Name=protonvpn-random
Comment=Automatically changes Proton VPN CLI connections
Exec=/usr/local/bin/protonvpn-randomizer
Terminal=true
StartupNotify=false
```

Terminal setting

Terminal=true

Opens a terminal window. The countdown and tput-based output are visible.

Terminal=false

Runs without a terminal window. Terminal-only messages and the countdown are not visible.

IMPORTANT SYNTAX

A desktop entry must contain either Terminal=true or Terminal=false. The literal value Terminal=true/false is not valid.

Verification

- Log out and back in, or launch the .desktop file manually.
- Confirm that the command path exists and is executable.
- Check desktop notifications or the terminal output depending on the selected mode.

Exit codes and troubleshooting

Use the process exit status to distinguish normal termination, missing dependencies and invalid configuration.

Defined exit codes

0	Normal exit.
1	General failure.
10	Required dependency is missing.
11	The configuration is invalid.

Display the last exit code

```
echo "$?"
```

IMMEDIATE CHECK

Run `echo "$?"` directly after `protonvpn-random` exits. Any intervening command replaces the stored exit status.

Troubleshooting sequence

1	Run <code>protonvpn-random -h</code> and verify the command is the expected script.
2	Confirm <code>bash</code> , <code>protonvpn</code> , <code>notify-send</code> and <code>tput</code> are available in <code>PATH</code> .
3	Run <code>protonvpn</code> directly to confirm login and location support.
4	Check <code>MODE</code> , probability ranges and the minimum/maximum switch interval.
5	Review whether the selected destinations support Secure Core.

Diagnostic commands

```
command -v bash protonvpn notify-send tput
protonvpn status
protonvpn config list
```

Kill switch and security

Use cautious wording: server switching is normally protected, but security claims depend on the active CLI and system configuration.

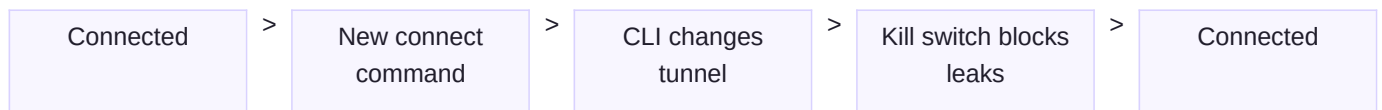
What Proton documents

Proton states that its kill switch blocks traffic when the VPN connection is lost and that, in most cases, it also protects while switching between VPN servers. This is not an unconditional guarantee for every platform, version or local network configuration.

Kill-switch-Guard

protonvpn-random will always force kill-switch standard (on)

Protection model during a switch



MAXIMUM ASSURANCE

Test the actual behavior on the target system before relying on unattended server changes. For stricter protection, consider a separately managed persistent firewall or kill-switch policy that does not depend solely on the script lifecycle.

Important distinction

protonvpn-random delegates connection changes to the Proton VPN CLI. The script itself does not implement packet filtering, firewall rules or a kill switch.

Compatibility and sources

Keep the manual useful as both protonvpn-random and the Proton VPN CLI evolve.

Compatibility boundary

The script depends on commands, options and destination identifiers provided by the installed Proton VPN CLI. A future CLI update may change behavior or require an update to protonvpn-random.

Recommended maintenance checks

- Run protonvpn-random -h after updating the script.
- Run protonvpn config list after updating the Proton VPN CLI.
- Retest city, country and server identifiers used in RANDOM_POOL.
- Retest kill-switch behavior after significant network or operating-system changes.

Official documentation checked

Proton VPN Linux CLI
<https://protonvpn.com/support/linux-cli>

Proton VPN kill switch
<https://protonvpn.com/support/what-is-kill-switch>

Document decisions

Structure	The new MODE variable is documented as its own selection layer rather than being hidden inside the random-pool section.
Duplication	The manual contains the full configuration reference; the separate USAGE.md contains a shorter standalone overview.
Corrections	Invalid typographic quotes, mixed-language fragments, grammar errors and the invalid Terminal=true/false example were normalized.
Security wording	Absolute leak-prevention claims were replaced with the qualified wording used by current official documentation.

End of manual

Configuration and usage documentation for protonvpn-random.